LLM-Guided ANN Index Optimization for Efficient Human-Object Interaction Retrieval

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ABSTRACT

Multimodal vector database management systems expose joint configuration spaces where ANN index parameters and cross-modal reranking weights must be tuned against an expensive rebuild-and-query oracle costing 8–9 minutes per evaluation. Classical HPO methods fail structurally: grid and random search carry no memory across evaluations, while Bayesian TPE with univariate marginals cannot represent the conditional dependencies that arise when engine-type gating and search-depth–fusion-weight coupling render parameters inherently non-separable; Gaussian Process surrogates (GP-BO) partially address this but remain sensitive to the non-stationarity introduced by categorical engine transitions. We present a phase-aware LLM optimization agent that compresses the full empirical trajectory $\mathcal{H}_t$ of (configuration, Score) pairs into structured prompts, enabling the model to reason over cross-stage dependencies and allocate its 30-iteration budget across exploration, exploitation, and refinement phases without a surrogate model. On HICO-DET — a 47,776-image, 600-category HOI retrieval benchmark with a 7-parameter coupled configuration space — the agent achieves mean Score = 4.44 (mAP × QPS), outperforming Optuna TPE and Random Search by +9.7% and GP-BO by +3.3%, reaching Score ≥ 3.97 at iteration 6 versus iteration 27 for Optuna TPE and iteration 28 for Random Search (≈2.7 hours of compute saved). Across two generalization benchmarks, performance gaps compress proportionally with coupling complexity — $\lesssim$1.7% spread on GLDv2 (762K images) and 2.1% on SIFT1M, where the convergence ordering inverts to favor Optuna TPE — validating that the agent's advantage scales with configuration-landscape coupling.

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1 INTRODUCTION

Modern vector database management systems expose a multidimensional configuration space whose parameters jointly govern the accuracy–throughput trade-off at production scale. A system such as VDMS [32], Milvus [33], or GaussDB-Vector [30] serving semantic similarity queries over tens of millions of embeddings requires setting graph connectivity (M), search depth (*efSearch*), candidate pool size (k), cross-modal fusion weight α, reranker reference count (n_{refs}), and reference selection strategy — a 7-parameter space in which each evaluation requires a full index rebuild and query execution across tens of thousands of embeddings, incurring significant wall-clock cost per trial [37] — making convergence speed a first-class system concern. The interactions among parameters are highly non-linear: the optimal value of each parameter depends on the values of the others in ways that resist independent analysis.

State-of-the-art approaches automate the search via grid enumeration, random sampling, or Bayesian optimization [3, 28]. These methods follow a trial-and-error strategy with no cross-parameter memory: grid and random search evaluate every candidate independently, while Bayesian methods such as Optuna TPE [1] model each parameter with independent marginal distributions [3]. This independence assumption is harmless on scalar parameter spaces, but it causes *structural* failure on multi-stage retrieval pipelines, where Stage-1 search depth (*efSearch*) and Stage-2 fusion weight (α) are jointly determined: reducing *efSearch* compresses the candidate pool fed to the reranker, an effect that can only be compensated by raising α. No parameter-independent technique models this inter-stage coupling, causing all three method families to systematically miss compensatory operating points — configurations that are globally optimal yet structurally invisible to blind optimizers regardless of iteration budget.

The practical severity of this failure is amplified by the cost structure of ANN index tuning. Each evaluation in a production-scale system requires a full index rebuild followed by query execution across tens of thousands of embeddings — 7–8 minutes of wall-clock time per iteration on representative hardware, capping a realistic tuning budget at approximately 30 iterations (3.5–4 hours per method). Under this constraint, a method that spends even 10–15 early iterations in suboptimal regions of the *efSearch*–α landscape does not merely converge slowly: it *fails* to locate the globally optimal configuration entirely. The LLM agent's ability to reason explicitly about inter-stage dependencies is therefore not a convenience — it is the mechanism that transforms a 30-iteration budget from insufficient to *sufficient* for navigating a coupled configuration landscape.

The core challenge is therefore: *how do we build an optimizer that reasons over the full history of (configuration, Score) pairs while respecting conditional parameter dependencies and cross-stage coupling?* A necessary precondition is a benchmark with sufficient coupling complexity to distinguish optimizer sophistication from random luck. Pure vector search datasets such as SIFT1M [11] and ANN-Benchmarks [2] expose a small, near-independent parameter space where any competent optimizer converges within tens of iterations. We address this gap by constructing the first text-query image retrieval benchmark derived from HICO-DET [5] — 47,776 images across 600 fine-grained verb–object interaction categories, deployed on Intel's VDMS vector DBMS — whose compositional structure generates a retrieval workload rich enough to expose the structural limits of blind ANN optimizers. Every prior use of HICO-DET has been restricted to the detection paradigm [13, 17, 24, 39]; we identify and fill the orthogonal retrieval direction. The compositional structure of the 600 HOI categories — spanning fine-grained verb–object pairings such as *ride bicycle*, *hold book*, and *pet cat* — creates a retrieval workload qualitatively harder than scene-level or object-level benchmarks: a successful query must locate images matching a specific *interaction*, not merely the presence of the involved entities. This structural complexity translates directly into a more sensitive configuration landscape, where the optimal *efSearch*–α trade-off differs substantially from that on simpler datasets, and where even exhaustive exact-search with modality fusion reaches only mAP = 0.276 — confirming that performance differences across optimizer methods reflect optimizer quality rather than dataset saturation. To evaluate on this benchmark we further construct a two-stage retrieval pipeline pairing CLIP ViT-L/14 ANN retrieval with zero-cost DINOv2 ViT-L/14-reg4 pseudo-relevance-feedback reranking, and formally prove that CLIP text embeddings occupy the semantic centroid of each HOI category's visual cluster — a geometric result that makes text queries structurally superior to image queries on compositionally structured data and directly justifies the reranker design without a secondary index.

Two parallel threads of prior work address orthogonal aspects of this problem, yet neither reaches the intersection our setting requires. On the *LLM-guided tuning* side, GPTuner [14], λ-Tune [10] (SIGMOD 2025), and OtterTune [31] show that LLM reasoning and learning-based methods outperform Bayesian and reinforcement-learning baselines on DBMS configuration — but all three operate on flat, scalar knob spaces (PostgreSQL, MySQL) where parameters are conditionally independent and no cross-stage coupling exists by construction. On the *ANN parameter tuning* side, VDTuner [37] applies multi-objective Bayesian optimization to VDMS index parameters, and VSAG [41] (VLDB 2025) eliminates index rebuilds via analytical subgraph monotonicity — but both treat parameters independently and neither models the coupling between *efSearch* and α that emerges when ANN retrieval feeds a reranker. Our work occupies the intersection that neither thread reaches: the first LLM-guided optimizer designed specifically for ANN index spaces with hard conditional type-dependencies and cross-stage coupling, reasoning over the joint efSearch–α trade-off that all parameter-independent methods structurally cannot model, regardless of iteration budget or LLM capability.

To address these challenges, we propose a *phase-aware LLM agent* that conditions each configuration proposal on the complete history of measured (configuration, Score) pairs, reasoning explicitly over cross-stage parameter dependencies across three self-regulated optimization phases — exploration, exploitation, and fine-tuning — with stagnation detection and anti-collapse mechanisms. The agent is deployed on Intel's VDMS and evaluated across three datasets of increasing parameter-space simplicity: HICO-DET (7-parameter, cross-modal, full cross-stage coupling), Google Landmarks v2 [35] (GLDv2, 762K images, image-to-image retrieval), and SIFT1M [11] (4-parameter, pure ANN search). Compared to state-of-the-art baselines, the agent achieves +9.7% accuracy–throughput Score over Optuna TPE and +3.3% over GP-BO (a VDTuner-style Gaussian Process surrogate) on HICO-DET, converging in ∼13 iterations on average versus ∼20 for Optuna TPE — a 2.6× improvement over the CLIP FaissFlat baseline. On GLDv2, the agent reaches near-optimal quality in ∼1–2 iterations versus ∼13 for Optuna TPE. On SIFT1M, all methods converge within 2% of each other — confirming that the agent's advantage scales with the structural coupling of the configuration landscape.

Contributions. This paper makes the following contributions:

(1) A phase-aware LLM agent for ANN hyperparameter optimization. We identify that ANN retrieval search spaces have hard conditional dependencies — engine type gates valid parameters, and Stage-1 search depth couples to Stage-2 fusion weight — that cause parameter-independent methods to fail structurally on this class of problems. We present a phase-aware LLM agent that conditions each proposal on the full history of (configuration, Score) pairs, discovering the compensatory operating point efSearch=64 < k=200 with α=0.80 that neither Bayesian nor random methods locate within 30 iterations. The agent converges to ∼95% of its best Score in ∼13 iterations on average versus ∼20 for Optuna TPE, and achieves a mean Score = 4.44 (mAP×QPS) across seeds — a 2.6× improvement over the CLIP FaissFlat baseline and 13.9× over brute-force CLIP+DINOv2 reranking. Mechanistic ablations confirm that *both* components are independently necessary: removing history conditioning degrades mean Score by 3.2% and removing phase-aware transitions by 5.2%, with neither ablation locating the compensatory efSearch=64, α=0.80 operating point across any seed.

(2) The first HOI retrieval benchmark. We construct a text-query image retrieval benchmark from HICO-DET (47,776 images, 600 compositional HOI categories, 90,641 ground-truth pairs), deployed on Intel's VDMS, providing a coupled configuration landscape rich enough to distinguish optimizer sophistication from random luck. The benchmark's fine-grained verb–object structure creates a retrieval task qualitatively harder than scene- or object-level benchmarks: even exhaustive exact-search with modality fusion reaches only mAP = 0.276, exposing a semantic ceiling that reflects language–vision alignment difficulty rather than any indexing limitation.

(3) A deployment-efficient two-stage retrieval pipeline. We propose pairing CLIP ViT-L/14 ANN retrieval with DINOv2 ViT-L/14-reg4 pseudo-relevance-feedback reranking at zero additional indexing cost, achieving +3.65 pp mAP (+15.2% relative) over ANN-only retrieval at under 1% additional query latency, with a single ANN index at deployment.

(4) A geometric theory of text query superiority. We formally prove that a CLIP text embedding occupies the semantic centroid of its HOI category's visual cluster, while any reference image is displaced from that centroid by $\varepsilon_j > 0$, contributing a bias term ε_j^2 to the expected distance from every class member. This centroid-displacement theory justifies the zero-cost reranker design — averaging n_r centroid-proximal reference images minimizes ε_j without a secondary index — and predicts the systematic mAP advantage of text queries over image queries on compositionally structured datasets.

(5) A complexity-matched evaluation across three retrieval workloads. We evaluate across HICO-DET, GLDv2, and SIFT1M — datasets of systematically increasing parameter-space simplicity — to characterize when and why LLM-guided reasoning outperforms classical methods. On HICO-DET the agent achieves +9.7% over Optuna TPE and +3.3% over GP-BO (VDTuner-style); on GLDv2 it reaches near-optimal quality in ∼1–2 iterations versus ∼13 for Optuna TPE; on SIFT1M all methods converge within 2%. This graded evaluation establishes that the agent's advantage is proportional to the structural coupling of the configuration landscape, providing a principled framework for predicting optimizer behavior across retrieval workloads.

The remainder of this paper is organized as follows. Section 2 reviews related work on HOI understanding, image retrieval, vector database systems, and hyperparameter optimization. Section 3 describes the construction of the HICO-DET retrieval benchmark. Section 4 presents the two-stage retrieval pipeline and the embedding geometry analysis. Section 5 details the LLM-guided ANN hyperparameter optimizer. Section 6 reports experimental results and ablations. Section 7 concludes.

2 RELATED WORK

2.1 Human-Object Interaction Detection

HOI detection has been studied extensively since the introduction of HICO-DET [5], which defined the two-stage paradigm of detecting human–object pairs and classifying their interactions across 600 fine-grained verb–object categories. Early approaches modeled spatial and appearance relationships between detected instances via instance-centric attention [9]. The adoption of transformer architectures enabled end-to-end detection [13, 39], and subsequent work leveraged vision-language pre-training to transfer interaction semantics from large-scale models [17, 18, 24]. HOICLIP [24] (CVPR 2023) distills CLIP interaction semantics into a detector via knowledge transfer, while DP-HOI [17] (CVPR 2024) applies disentangled pre-training to reduce annotation dependence. Despite sustained progress on mean Average Precision for detection, every published use of HICO-DET is restricted to the detection paradigm — predicting bounding boxes and interaction labels from a given image. We identify and fill the orthogonal *retrieval* direction: given a natural-language HOI query, rank all 47,776 database images by relevance. The compositional verb–object label structure that makes HOI detection challenging creates an equally demanding retrieval workload, one with a 7-parameter coupled ANN configuration space that exposes the structural limits of blind optimizers.

2.2 Cross-Modal Image Retrieval

Text-image retrieval research established benchmark datasets from caption annotations [19, 38] and proposed contrastive embedding losses to bring matching pairs close in a shared space [6]. CLIP [26] transformed the field: its joint vision-language embedding enables zero-shot retrieval across diverse domains without task-specific fine-tuning. BLIP-2 [15] extended this paradigm by bridging frozen visual encoders and large language models via a lightweight querying transformer, achieving strong generalization with substantially fewer trainable parameters. UniIR [34] (ECCV 2024 Oral) unified eight multimodal retrieval tasks in a single instruction-tuned model, constructing the M-BEIR benchmark spanning 1.5M queries over 10 datasets; we adopt UniIR and standalone CLIP as primary baselines. LamRA [21] (CVPR 2025) repurposes generative large multimodal models for universal retrieval via lightweight LoRA adaptation, achieving strong cross-task generalization on M-BEIR without task-specific fine-tuning. None of these works constructs a retrieval benchmark from HOI annotations, analyzes the relative merits of text versus image queries on compositionally structured data, or addresses the downstream ANN index configuration problem that arises when deploying such a retrieval system in a production VDMS. We fill all three gaps.

2.3 Vector Database Management Systems and ANN Indexing

ANN index research has produced a rich family of algorithms: graph-based indices (HNSW [22], DiskANN [29]), inverted file structures with product quantization [11], and GPU-accelerated flat search [12]. HNSW [22] constructs a hierarchical proximity graph supporting $O(\log n)$-complexity search with tunable accuracy–throughput trade-offs governed by M and *efSearch*; DiskANN [29] extends billion-scale graph search to SSD-resident indices. ANN-Benchmarks [2] and the NeurIPS 2021 billion-scale competition [27] systematically characterize the Pareto frontier of recall versus throughput across these algorithms and datasets.

Production VDBMSs integrate ANN indexing with metadata management, persistent storage, and query planning. Milvus [33] supports multiple index backends with a unified query interface; VDMS [32] co-locates PMGD graph-property metadata with vector data, enabling structured pre-filters on ANN queries — a feature we exploit as one optimization dimension. GaussDB-Vector [30] (VLDB 2025) decouples indexing from serving for billion-vector scale. VSAG [41] (VLDB 2025) eliminates costly rebuilds during parameter sweeps via subgraph monotonicity. Despite this infrastructure richness, *index configuration* — choosing engine type, graph connectivity M, search depth *efSearch*, candidate pool size k, and cross-stage fusion weight α for a given embedding distribution and throughput target — is universally left to blind grid or random search. Our LLM agent replaces these procedures with history-conditioned sequential reasoning over the full mixed-type parameter space.

2.4 Hyperparameter Optimization

HPO methods divide into three families. **Heuristic search:** Grid and random search [3] are the dominant baselines; random search is empirically superior in high-dimensional spaces because it covers more of the marginal space of each parameter. **Bayesian optimization (BO):** Gaussian-process-based BO [28] fits a probabilistic surrogate and acquires the next configuration by maximizing expected improvement; BOHB [7] hybridizes kernel density estimation with successive halving for additional efficiency. **Bandit and meta-learning:** Hyperband [16] treats configuration selection as a pure-exploration bandit problem; Auto-sklearn [8] applies BO to joint algorithm selection and hyperparameter configuration. ML-guided DBMS configuration was pioneered by OtterTune [31], which applied Gaussian process regression to map workload features to optimal knob settings; subsequent systems applied reinforcement learning, but all operate on continuous or integer knob spaces without structural type-dependencies.

Most closely related on the VDMS side, VDTuner [37] applies multi-objective Bayesian optimization to tune both index parameters and system parameters in Milvus, using a successive-abandon strategy to allocate the evaluation budget across index types with non-fixed parameter spaces, achieving up to 3.57× faster convergence than baselines. However, VDTuner evaluates on pure vector search benchmarks (GloVe, keyword-match) without semantic structure and models all parameters independently via a Gaussian process surrogate — it does not account for the cross-stage coupling between ANN search depth and reranker fusion weight that arises in multi-stage retrieval pipelines.

The fundamental limitation of all classical HPO methods for ANN index selection is that the search space is *mixed-type*: discrete engine choices (*HNSW* vs. *IVFFlat*), integer parameters (M, *efSearch*, *nlist*, k), and continuous parameters (α), with hard structural dependencies between parameters and engine choice. Gaussian process surrogates assume stationarity over a continuous space, making them ill-suited to this setting. Tree-structured Parzen Estimators (TPE), as implemented in Optuna [1], handle mixed-type spaces more gracefully by modeling the density of good and bad configurations separately, but they treat each parameter independently and structurally cannot exploit the knowledge that reducing *efSearch* compresses the candidate pool fed to a reranker — an effect that must be compensated by raising α. This independence assumption causes TPE to miss the compensatory operating point efSearch=64, α=0.80 even within a 30-iteration budget.

2.5 LLM-Guided System Configuration

Recent work established that large language models can act as black-box optimizers by iteratively proposing candidate solutions and refining them based on observed outcomes. OPRO [36] encodes the history of (solution, score) pairs directly in the prompt, enabling the LLM to generate improved solutions without gradient information. Follow-up work applied this paradigm to machine learning hyperparameter optimization [20, 40], showing that LLMs match or surpass Bayesian optimization in sample efficiency on tabular HPO benchmarks by exploiting domain knowledge embedded in pre-training.

Table 1: Comparison of closely related configuration optimization methods. ✓/× indicate whether the property is present.

Method	LLM-Guided	Cross-Stage Coupling	Mixed-Type Space	Semantic Benchmark
OtterTune [31]	×	×	×	×
GPTuner [14]	✓	×	×	×
λ-Tune [10]	✓	×	×	×
VDTuner [37]	×	×	✓	×
Optuna TPE [1]	×	×	✓	×
Ours	✓	✓	✓	✓

In the database configuration domain, GPTuner [14] (VLDB 2024) uses GPT-4 to extract structured priors from DBMS manuals and guide a coarse-to-fine Bayesian optimization of PostgreSQL and MySQL knobs, building on OtterTune [31] with substantially reduced tuning cost. λ-Tune [10] (SIGMOD 2025) harnesses LLMs for automated database system tuning, using LLM-generated workload analysis to warm-start configuration search and demonstrating that LLM reasoning transfers to heterogeneous knob spaces. Both systems operate on flat, scalar knob spaces (buffer pool sizes, parallelism settings, I/O parameters) where parameters are conditionally independent by construction: no engine type gates valid parameters, and no cross-stage coupling arises because DBMS knob tuning has a single-objective, single-stage evaluation loop.

Our work occupies the intersection that neither the HPO thread (VDTuner [37], Optuna [1], random search [3]) nor the LLM-guided DBMS tuning thread (GPTuner [14], λ-Tune [10], OtterTune [31]) reaches: a history-conditioned LLM agent designed specifically for ANN index spaces with *hard conditional type-dependencies* (engine type gates valid parameters) and *cross-stage coupling* (*efSearch* and α must be jointly optimized when ANN retrieval feeds a reranker). Methods from the HPO thread are structurally incapable of modeling this coupling regardless of iteration budget; methods from the LLM-DBMS thread do not encounter it by construction. Mechanistic ablations confirm that history conditioning and phase-aware transitions are both independently necessary to navigate this coupled space: removing either prevents the agent from locating the compensatory efSearch=64, α=0.80 operating point across all seeds. To our knowledge, this is the first application of LLM-guided optimization to ANN index architecture selection with cross-stage parameter coupling in a vector database management system.

3 THE HICO-DET RETRIEVAL BENCHMARK

3.1 Source Dataset

HICO-DET [5] contains 47,776 images (38,118 train / 9,658 test) annotated across 600 HOI categories with per-instance bounding boxes and interaction labels, where each category is a *verb–object* pair drawn from 117 action verbs and 80 MS-COCO object classes. Three structural properties make HICO-DET distinctively demanding as a benchmark for ANN index configuration optimization. **P1 — Compositional fine-grained discrimination.** The 600 categories span 80 COCO object classes and 117 verb types; categories sharing the same object (e.g., *ride horse*, *sit_on horse*, *straddle horse*) differ only in the action verb. Single-encoder retrieval conflates these closely related categories, making the joint selection of ANN engine and Stage-2 fusion weight critical to performance. **P2 —**

Long-tail relevance. Category sizes are highly skewed: the largest categories exceed 1,000 positive images while the smallest have as few as 5. An optimizer that maximizes mean mAP without attending to per-category variance will exploit high-frequency categories while failing on rare but equally weighted queries. **P3 — Dual-encoder modality gap.** CLIP retrieves by language-aligned semantic similarity; DINOv2 captures fine-grained visual structure complementary to CLIP's representations [25]. Neither encoder alone achieves optimal mAP@10, making the Stage-2 fusion weight α a data-dependent parameter that cannot be fixed by hand and must be tuned jointly with the Stage-1 ANN configuration.

Every published use of HICO-DET has been restricted to the detection paradigm [13, 17, 24, 39]: given an image, predict bounding boxes and interaction class labels. We repurpose these rich compositional annotations to construct, to our knowledge, the first text-query image retrieval benchmark from HICO-DET, opening an orthogonal evaluation axis that the detection literature has not explored.

3.2 Benchmark Construction

Retrieval Corpus: Why Merge Train and Test Splits. We use all 47,776 images as a single retrieval corpus. The original HICO-DET train/test split was designed for supervised detection training and carries no meaning for retrieval evaluation, where the task is to rank corpus images given a query, not to generalize across a detection boundary. Merging both splits maximizes the positive-pair density needed for meaningful mAP computation across 600 categories; restricting evaluation to the 9,658-image test split alone would reduce per-category positive counts below the minimum threshold for reliable mAP@10 computation across all 600 classes. CLIP ViT-L/14 [26] embeddings are pre-computed offline using the official OpenAI weights and stored in VDMS as a 768-dimensional L2 DescriptorSet of 47,776 vectors.

Query Design: One Canonical Query per HOI Class. We construct exactly one canonical natural-language text query per HOI category, yielding 600 queries. A single canonical query per category eliminates inter-query variance from paraphrase choice and ensures that each of the 600 HICO-DET categories contributes equally to the mean Average Precision score — a design decision motivated by the same reproducibility principle used in standard IR evaluation [23]. Each query follows the template *"a person {verb} a {object}"*, with the special case *"a person near a {object}"* for the $\mathrm{no\_interaction}$ verb class. Queries are encoded with the same CLIP ViT-L/14 text encoder, producing 768-dimensional unit-norm text embeddings that reside in the same embedding space as the indexed image vectors.

Ground Truth: Multi-Label Positive Structure. An image is relevant to a query if it carries the corresponding HOI annotation in the HICO-DET ground truth. A single image may be relevant to multiple queries (e.g., an image annotated with both *ride motorcycle* and *hold phone* contributes positives to two queries). Summed over all 600 queries, there are 90,641 positive (image, query) pairs. Every query has at least 5 relevant images, so mAP@10 is defined and non-trivially bounded for each of the 600 queries.

3.3 Evaluation Protocol

We evaluate retrieval quality using mean Average Precision at k=10 (mAP@10), Precision@10, NDCG@10, and Recall@10 over all 600 queries. mAP@10 is our primary metric: it rewards systems that place relevant images in the top-10 positions and is the standard metric for ranked retrieval with multiple relevant documents per query [23]. Throughput is measured in queries per second (QPS), computed as 600 divided by the total wall-clock time for all queries issued as a single batch to a live VDMS container. Batch QPS is the appropriate production metric: a deployed HOI retrieval system serves queries at steady-state rates that are best captured by batch throughput; single-query latency at this scale is dominated by container and network overhead rather than by the ANN index structure, which would mask precisely the index differences the optimizer is designed to exploit. The joint optimization objective is

$$\text{Score} = \text{mAP} \times \text{QPS}. \tag{1}$$

This scalar encodes the accuracy–throughput trade-off symmetrically: gains in QPS and gains in mAP contribute multiplicatively, so neither axis can be sacrificed without penalizing the objective. Section 3.4 below formalizes the configuration space and the optimization problem whose objective is this Score.

3.4 Problem Statement

Definition 1 (Configuration Space)**.** Let Θ denote the joint configuration space of the two-stage pipeline (Section 4). Θ is the Cartesian product of seven parameter domains (*engine*; M, nlist; efSearch, nprobe; k; α; n_{refs}; $\mathrm{ref\_strategy}$) enumerated in Table 2; $|\Theta| = 102{,}144$.

Definition 2 (Evaluation Oracle)**.** For any configuration $\theta \in \Theta$, the evaluation oracle $O(\theta) = \big(\mathrm{mAP}(\theta),\ \mathrm{QPS}(\theta)\big)$ is obtained by (i) rebuilding the VDMS ANN index under θ, and (ii) issuing all 600 HICO-DET text queries to the live container and recording mAP (Section 3) and batch QPS. The oracle is *black-box*: no closed-form gradient with respect to θ is available, and each call requires a full index rebuild followed by 600 retrieval operations.

Definition 3 (Optimization Problem)**.** Given Θ (Definition 1) and O (Definition 2), find

$$\theta^* \;=\; \arg\max_{\theta\in\Theta}\ \mathrm{Score}(\theta), \tag{2}$$

where $\mathrm{Score}(\theta) = \mathrm{mAP}(\theta) \times \mathrm{QPS}(\theta)$ (Eq. (1)).

Intractability. Exhaustive search over $|\Theta| = 102{,}144$ configurations is infeasible under any practical budget: each oracle call requires 3–10 minutes of wall-clock time (index rebuild plus 600 query evaluations on a GPU node), so evaluating the full space would require thousands of GPU-hours. At the N=30-iteration budget used in our study, a non-adaptive strategy covers at most 0.03% of Θ; evaluation feedback is additionally noisy due to VDMS runtime variability. This combination — a black-box, expensive oracle with an exponential search space and a tiny sample budget — is precisely the regime where LLM-guided reasoning offers the greatest advantage over grid and random baselines.

Cross-Stage Parameter Coupling. The seven parameters decompose into two coupled groups: Stage-1 parameters {engine, M, $\mathrm{efSearch}$,

Table 2: Configuration space Θ (Definition 1). †Conditional on engine. Total: HNSW $5{\times}7{\times}8{\times}19{\times}4{\times}3{=}63{,}840$**; IVF** $4{\times}5{\times}8{\times}19{\times}4{\times}3{=}36{,}480$**; Flat** $8{\times}19{\times}4{\times}3{=}1{,}824$**.**

Parameter	**Notation**	**Value Set**	**Card.**
Stage 1 — ANN retrieval			
engine	—	Flat, HNSWFlat, IVFFlat	3
conn. depth† (HNSW)	M	8, 16, 32, 48, 64	5
cluster count† (IVF)	nlist	64, 128, 256, 512	4
search beam† (HNSW)	efSearch	32–500 (7 vals)	7
probed cells† (IVF)	nprobe	4, 8, 16, 32, 64	5
candidate pool	k	50–1000 (8 vals)	8
Stage 2 — DINOv2 reranking			
fusion weight	α	0.0, 0.05, ..., 0.90	19
ref. count	n_{refs}	1, 3, 5, 10	4
ref. strategy	—	first, centroid, diverse	3
Total $\lvert\Theta\rvert$			**102,144**

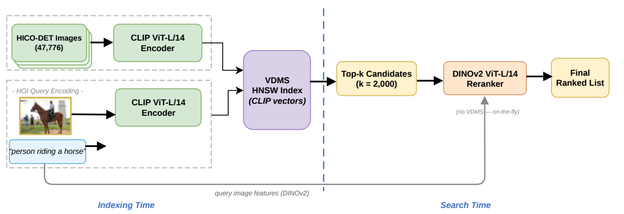

Figure 1: Two-stage retrieval pipeline (oracle $\mathcal{O}(\theta)$). Stage 1: CLIP ViT-L/14 ANN retrieval produces a top-k candidate pool. Stage 2: DINOv2 ViT-L/14-reg4 distances are fused via convex combination $s_j{=}(1{-}\alpha)\hat{c}_j + \alpha\hat{d}_j$, re-ranking candidates at zero additional indexing cost. Score $= \mathrm{mAP} \times \mathrm{QPS}$**.**

nprobe, $k\}$ govern ANN recall and throughput, while Stage-2 parameters $\{\alpha, n_{\text{refs}}, \text{ref\_strategy}\}$ govern DINOv2 reranking. The optimal pool size k and fusion weight α are *jointly determined*: at small k, Stage 1 returns only high-confidence CLIP candidates so DINOv2 reranking adds marginally, favouring small α; at large k, the pool contains visually diverse images that benefit from DINOv2 reranking, favouring larger α. This structural coupling invalidates the parameter-independence assumption underlying GP-based Bayesian methods [37] and Optuna TPE [1], which model each parameter's marginal contribution independently and cannot represent cross-stage interactions. Our LLM agent explicitly reasons about these inter-parameter dependencies in its natural-language analysis of prior outcomes (Section 5).

Section Outline. Section 4 constructs the two-stage pipeline that implements oracle $\mathcal{O}$, mapping any $\theta \in \Theta$ to a $(\mathrm{mAP}, \mathrm{QPS})$ pair and formally enumerating all parameter domains. Section 5 presents the LLM-guided optimizer that solves $\arg\max_{\theta\in\Theta} \mathrm{Score}(\theta)$ adaptively within N=30 oracle evaluations. Section 6 validates the solution on HICO-DET, GLDv2, and SIFT1M, reporting mean Score across three random seeds against five optimization baselines.

4 TWO-STAGE RETRIEVAL PIPELINE

4.1 Oracle Instantiation and Notation

Oracle $\mathcal{O}$ (Definition 2) maps any $\theta \in \Theta$ to a $(\mathrm{mAP}, \mathrm{QPS})$ pair; the optimizer maximizes $\mathrm{Score}(\theta) = \mathrm{mAP}(\theta) \times \mathrm{QPS}(\theta)$. Let $C = \{(\mathrm{c}_i, \mathrm{d}_i)\}_{i=1}^{N}$ denote the corpus of N=47,776 images, where $\mathrm{c}_i \in \mathbb{R}^{768}$ is the unit-normalized CLIP ViT-L/14 embedding and $\mathrm{d}_i \in \mathbb{R}^{1024}$ is the DINOv2 ViT-L/14-reg4 embedding of image i, and $\mathcal{I}_h \subseteq [N]$ the annotated relevant set for HOI class h. Given a text query q_h (e.g., *"a person riding a horse"*), the pipeline proceeds in two stages as illustrated in Figure 1.

Stage 1 encodes q_h into a unit-norm vector $\mathrm{q}_c \in \mathbb{R}^{768}$ via the frozen CLIP ViT-L/14 text encoder and issues an ANN search over the VDMS CLIP DescriptorSet to retrieve a candidate pool $\mathcal{T}_k(q_h)$ of cardinality k. **Stage 2** reranks $\mathcal{T}_k(q_h)$ using DINOv2 visual distances computed against a pre-built per-class reference r_h, fusing both distance signals without any additional index. The seven parameters of θ — engine, M or nlist, efSearch or nprobe, k, α, n_{refs}, ref_strategy — govern the accuracy–throughput trade-off across both stages and constitute the search space Θ (Table 2). The oracle cost per call is dominated by index construction ($\approx$425–455 s) and 600 batch queries ($\approx$35–99 s), yielding $\approx$8–9 minutes of wall-clock time per oracle evaluation.

4.2 Stage 1: ANN Candidate Generation

All N CLIP ViT-L/14 image embeddings are pre-computed offline and stored in VDMS as a 768-dimensional L2 DescriptorSet. At query time, q_h is encoded by the frozen text encoder and submitted to VDMS as a FindDescriptor request. VDMS returns the top-k candidate identifiers together with their L2 distances $\delta_j^c = \|\mathrm{c}_j - \mathrm{q}_c\|_2$; the candidate pool is formally

$$\mathcal{T}_k(q_h) = \{ j \in [N] : \|\mathrm{c}_j - \mathrm{q}_c\|_2 \le \delta^{(k)} \}, \tag{3}$$

where $\delta^{(k)}$ denotes the k-th smallest L2 distance from q_c to any corpus embedding. For FaissFlat, Eq. (3) is satisfied exactly; for FaissHNSWFlat and FaissIVFFlat, VDMS returns a high-recall approximation whose tightness is governed by the engine-specific search parameters.

Three ANN engines expose qualitatively distinct accuracy–throughput operating points:

- **FaissFlat**: exact exhaustive L2 search over all N embeddings; provides the recall upper bound for Stage 1 at the cost of the lowest QPS.
- **FaissHNSWFlat** [22]: hierarchical navigable small-world graph index; recall and latency are jointly controlled by graph connectivity $M \in \{8, 16, 32, 48, 64\}$ and beam-search width $\mathrm{efSearch} \in \{32, 64, 100, 150, 200, 300, 500\}$. Higher M increases graph density (larger index, better recall per beam step); higher efSearch expands the candidate frontier (better recall, lower QPS).
- **FaissIVFFlat** [12]: inverted-file index partitioning the embedding space into $\mathrm{nlist} \in \{64, 128, 256, 512\}$ Voronoi cells; search probes $\mathrm{nprobe} \in \{4, 8, 16, 32, 64\}$ cells. Larger nlist increases partition granularity; larger nprobe increases recall at the cost of proportionally higher latency.

The candidate pool size $k \in \{50, 100, 150, 200, 300, 500, 750, 1000\}$ determines the recall ceiling available to Stage 2 and is selected jointly with engine parameters. Batch QPS is defined as

$$\mathrm{QPS}(\theta) = \frac{|Q|}{T_{\text{total}}(\theta)}, \tag{4}$$

where $|Q|$=600 is the number of HOI text queries and $T_{\text{total}}(\theta)$ is the elapsed wall-clock time for all queries issued in a single batch to the live VDMS container under configuration θ. This end-to-end latency encompasses the VDMS round-trip for Stage 1, the in-memory reranking computation in Stage 2, and all inter-process communication overhead.

4.3 Stage 2: Feature Fusion and Reranking

Stage 2 reranks $\mathcal{T}_k(q_h)$ using DINOv2 ViT-L/14-reg4 [25] visual features — capturing fine-grained appearance structure complementary to CLIP's language-aligned semantics — without constructing any additional index. Each HOI class h is represented by a visual reference vector $\mathrm{r}_h \in \mathbb{R}^{1024}$, built offline as the mean DINOv2 embedding of a selected subset $\mathcal{S}_h \subseteq \mathcal{I}_h$:

$$\mathrm{r}_h = \frac{1}{|\mathcal{S}_h|} \sum_{i \in \mathcal{S}_h} \mathrm{d}_i, \tag{5}$$

where $n_r = |\mathcal{S}_h| \in \{1, 3, 5, 10\}$. For n_r=1 (the default), r_h is the DINOv2 embedding of the first image in the HOI category. For n_r>1, three strategies populate $\mathcal{S}_h$: **first** takes the initial n_r images in dataset order; **centroid** selects the n_r images nearest to the DINOv2 class mean $\boldsymbol{\mu}_h^d = \frac{1}{|\mathcal{I}_h|} \sum_{i \in \mathcal{I}_h} \mathrm{d}_i$, anchoring r_h at the visual core of the class; **diverse** applies greedy farthest-point sampling from the most central image. Reference construction requires <1 ms per class and does not contribute to T_{total} in Eq. (4).

CLIP distances $\delta_j^c = \|\mathrm{c}_j - \mathrm{q}_c\|_2$ and DINOv2 distances $\delta_j^d = \|\mathrm{d}_j - \mathrm{r}_h\|_2$ operate at different absolute scales; both are min-max normalized within $\mathcal{T}_k(q_h)$ onto [0, 1]:

$$\hat{\delta}_j^m = \frac{\delta_j^m - \min_{i \in \mathcal{T}_k} \delta_i^m}{\max_{i \in \mathcal{T}_k} \delta_i^m - \min_{i \in \mathcal{T}_k} \delta_i^m + \varepsilon}, \quad m \in \{c, d\}, \tag{6}$$

where $\varepsilon = 10^{-10}$, $\hat{c}_j \equiv \hat{\delta}_j^c$, and $\hat{d}_j \equiv \hat{\delta}_j^d$. Pool-local normalization preserves the fine-grained relative ordering of CLIP distances within $\mathcal{T}_k(q_h)$: intra-pool CLIP L2 values span a narrow range (≈1.55–1.75 for unit-normalized 768-d embeddings), a signal that a global corpus statistic would collapse.

The fused relevance score for candidate $j \in \mathcal{T}_k(q_h)$ is the convex combination [4]

$$s_j = (1 - \alpha)\, \hat{c}_j + \alpha\, \hat{d}_j, \tag{7}$$

where $\alpha \in \{0.0, 0.05, \ldots, 0.90\}$ (19 values); lower s_j indicates higher predicted relevance. Setting α=0 degenerates to CLIP-only retrieval; α=1 to DINOv2-only reranking. The convex combination is preferred over reciprocal rank fusion because CLIP text queries on a unit-normalized 768-d space produce meaningful intra-pool distance values whose fine-grained ordering is destroyed when converted to ranks. The optimal α is configuration-dependent and is learned jointly with Stage 1 parameters by the LLM optimizer (Section 5).

4.4 Text Query Superiority: Centroid-Displacement Theory

Let $\boldsymbol{\mu}_h^c = \frac{1}{|\mathcal{I}_h|} \sum_{i \in \mathcal{I}_h} \mathrm{c}_i$ denote the visual centroid of HOI class h in the CLIP space, and let $\sigma_h^2 = \frac{1}{|\mathcal{I}_h|} \sum_{i \in \mathcal{I}_h} \|\mathrm{c}_i - \boldsymbol{\mu}_h^c\|_2^2$ denote the intra-class visual variance. For any query vector $\mathrm{q}' \in \mathbb{R}^{768}$, the bias–variance decomposition of the expected squared L2 distance to a class member c_i drawn uniformly from $\mathcal{I}_h$ yields

$$\mathbb{E}_i\left[\|\mathrm{q}' - \mathrm{c}_i\|_2^2\right] = \underbrace{\|\mathrm{q}' - \boldsymbol{\mu}_h^c\|_2^2}_{\text{centroid bias}^2} + \underbrace{\sigma_h^2}_{\text{intra-class variance}}. \tag{8}$$

The variance term σ_h^2 is an intrinsic property of the class; only the bias term $\|\mathrm{q}' - \boldsymbol{\mu}_h^c\|_2^2$ depends on the query.

CLIP's contrastive training objective aligns the text embedding of a category description with the visual embeddings of all matching images [26]. For a compositionally structured category — e.g., *"a person riding a horse"* spanning diverse viewpoints, backgrounds, and subjects — the text embedding aggregates distributional statistics from many matched images and therefore lies near the visual centroid: $\mathrm{q}_c \approx \boldsymbol{\mu}_h^c$. The centroid bias is thus approximately zero:

$$\mathbb{E}_i\left[\|\mathrm{q}_c - \mathrm{c}_i\|_2^2\right] \approx \sigma_h^2. \tag{9}$$

In contrast, any reference image c_j ($j \in \mathcal{I}_h$) lies at centroid displacement $\varepsilon_j = \|\mathrm{c}_j - \boldsymbol{\mu}_h^c\|_2 > 0$, incurring a strictly positive bias:

$$\mathbb{E}_i\left[\|\mathrm{c}_j - \mathrm{c}_i\|_2^2\right] = \varepsilon_j^2 + \sigma_h^2 > \sigma_h^2. \tag{10}$$

The text query therefore achieves strictly lower expected squared distance to every class member than any single reference image, by a margin of $\varepsilon_j^2 > 0$. This centroid-displacement advantage is monotonically amplified by intra-class visual diversity: the higher the variance σ_h^2 across HOI images (e.g., *ride horse* spans many settings and viewpoints), the larger the typical ε_j and hence the larger the margin over any individual reference image. Under L2-based ANN retrieval, lower expected squared distance to class members translates directly to higher expected recall at any fixed k, and therefore to higher mAP. This result formally justifies the design choice of text queries for Stage 1 and additionally motivates the n_r>1 reference aggregation in Stage 2: averaging n_r class embeddings reduces the expected L_2 centroid bias of r_h by $O(1/\sqrt{n_r})$ relative to a single reference image.

5 LLM-GUIDED ANN HYPERPARAMETER OPTIMIZATION

The pipeline of Section 4 exposes a seven-parameter joint configuration $\theta \in \Theta$: engine, two engine-conditional index parameters, pool size k, fusion weight α, reference count n_r, and reference strategy. Because engine choice renders some parameters inapplicable—M and efSearch are undefined for IVF; nlist and nprobe are undefined for HNSW—the joint space $|\Theta|$=102,144 is engine-conditional and mixed-type. Each evaluation requires a full VDMS rebuild plus a 600-query pass (≈8–9 min per oracle call), ruling out exhaustive or surrogate-heavy strategies that require thousands of evaluations. We address this with a sequential LLM optimizer that exploits structural knowledge of Θ to allocate its N-iteration budget efficiently.

5.1 Search Space and Coupling Structure

The configuration space decomposes as

$$\Theta = \Theta_{\text{HNSW}} \cup \Theta_{\text{IVF}} \cup \Theta_{\text{Flat}}, \tag{11}$$

where each component is defined by its engine-conditional index parameters together with shared parameters k, α, n_r, and ref_strategy

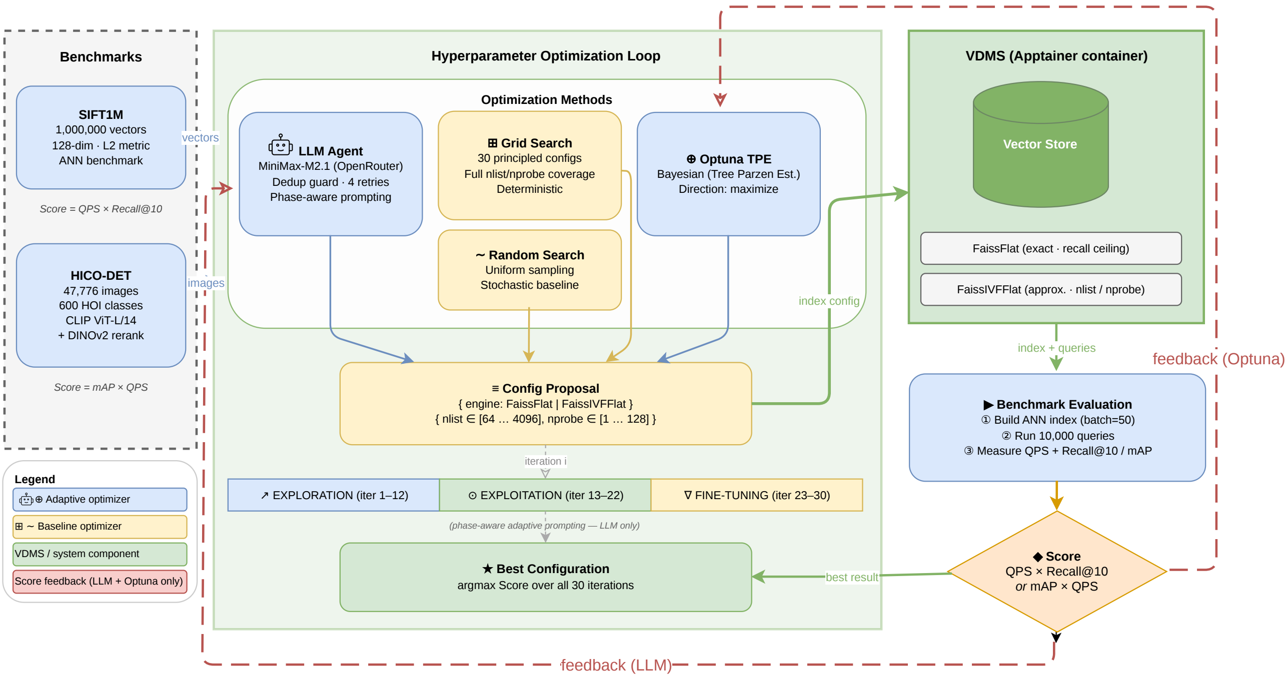

Figure 2: Optimization framework: five methods propose VDMS configurations over N=30 iterations. Adaptive methods (LLM Agent, Optuna TPE, GP-BO) receive score feedback (red arrows); Grid and Random Search do not.

(full value sets in Table 2). FaissHNSWFlat contributes M and efSearch; FaissIVFFlat contributes nlist and nprobe. The oracle $O(\theta)$ loads θ into a fresh VDMS instance, indexes the corpus, and evaluates all $|Q|$=600 HOI queries, returning the pair $(\mathrm{mAP}(\theta), \mathrm{QPS}(\theta))$ and $\mathrm{Score}(\theta) = \mathrm{mAP}(\theta)\times\mathrm{QPS}(\theta)$.

Two cross-parameter couplings invalidate the independence assumption shared by GP-BO [37] and Optuna TPE. First, pool size k determines the candidate set passed from Stage 1 to Stage 2; increasing k raises Stage 2 mAP recall but reduces throughput, so the joint (k, α) frontier is not separable: optimizing α at a suboptimal k cannot reach the Score achievable by joint search. Second, engine selection sets the QPS range available to any efSearch sweep, so structural decisions must precede continuous refinement. These coupled dependencies motivate a phase-partitioned budget that resolves structural choices before continuous ones.

5.2 History Trajectory and Prompt Generation

Let $h_t = (\theta_t,\ \mathrm{mAP}(\theta_t),\ \mathrm{QPS}(\theta_t))$ denote the evaluation record at iteration t and define the *search history trajectory*

$$\mathcal{H}_t \;=\; (h_1, h_2, \ldots, h_t), \quad t \geq 0, \quad \mathcal{H}_0 = \emptyset. \tag{12}$$

$\mathcal{H}_t$ is the agent's complete internal state: no surrogate model, gradient estimate, or kernel is maintained between iterations. At each iteration the agent receives a single structured prompt built from four components: (i) fixed system context encoding the VDMS pipeline architecture and per-engine parameter physics; (ii) the diagnostic class $g(\mathcal{H}_{t-1})$ (Section 5.2.1); (iii) the complete ordered history $\mathcal{H}_{t-1}$ as structured text; and (iv) dynamically computed untried-value hints (Section 5.2.2). The agent responds with a structured JSON object specifying the next θ_t and a *reasoning* field that articulates its rationale (Figure 3).

5.2.1 *Diagnostic Guidance Function.*

Let $\mathrm{Score}^*_t = \max_{s \leq t} \mathrm{Score}(\theta_s)$ denote the running best at the end of iteration t. The diagnostic class

$$g(\mathcal{H}_t) = \begin{cases} \textit{excellent} & \text{if } \mathrm{Score}(\theta_t) \;\geq\; 0.98\ \mathrm{Score}^*_t, \\ \textit{low\_mAP} & \text{if } \mathrm{mAP}(\theta_t) < 0.20 \text{ and } \mathrm{Score}(\theta_t) < 0.80\ \mathrm{Score}^*_t, \\ \textit{low\_QPS} & \text{if } \mathrm{mAP}(\theta_t) \geq 0.20 \text{ and } \mathrm{QPS}(\theta_t) < 4.0, \\ \textit{below\_best} & \text{if } \mathrm{Score}(\theta_t) < 0.90\ \mathrm{Score}^*_t, \\ \textit{moderate} & \text{otherwise}, \end{cases} \tag{13}$$

is injected into the prompt as a targeted natural-language directive. *Excellent* reinforces the current structural region and encourages incremental α or n_r refinement; *low_mAP* directs the agent to increase k or M; *low_QPS* directs it to reduce efSearch or switch engines; *below_best* encourages structural divergence; and *moderate* suggests coordinated adjustment of k and α.

5.2.2 *Untried-Value Hints and Duplicate Avoidance.*

During EXPLOITATION and FINE-TUNING, the prompt additionally provides $\mathcal{U}_t$: untried α values and $(n_r, \mathrm{ref\_strategy})$ pairs under the current best engine, derived from $\mathcal{H}_t$. These hints prevent the agent from reproposing already-evaluated continuous-parameter combinations once the structural choice is settled. Each proposed θ_{t+1} is checked against $\mathcal{H}_t$ before evaluation; if a duplicate is detected, the prompt is resubmitted with an explicit rejection notice until a novel configuration is accepted or a uniformly random fallback is applied, guaranteeing loop progress.

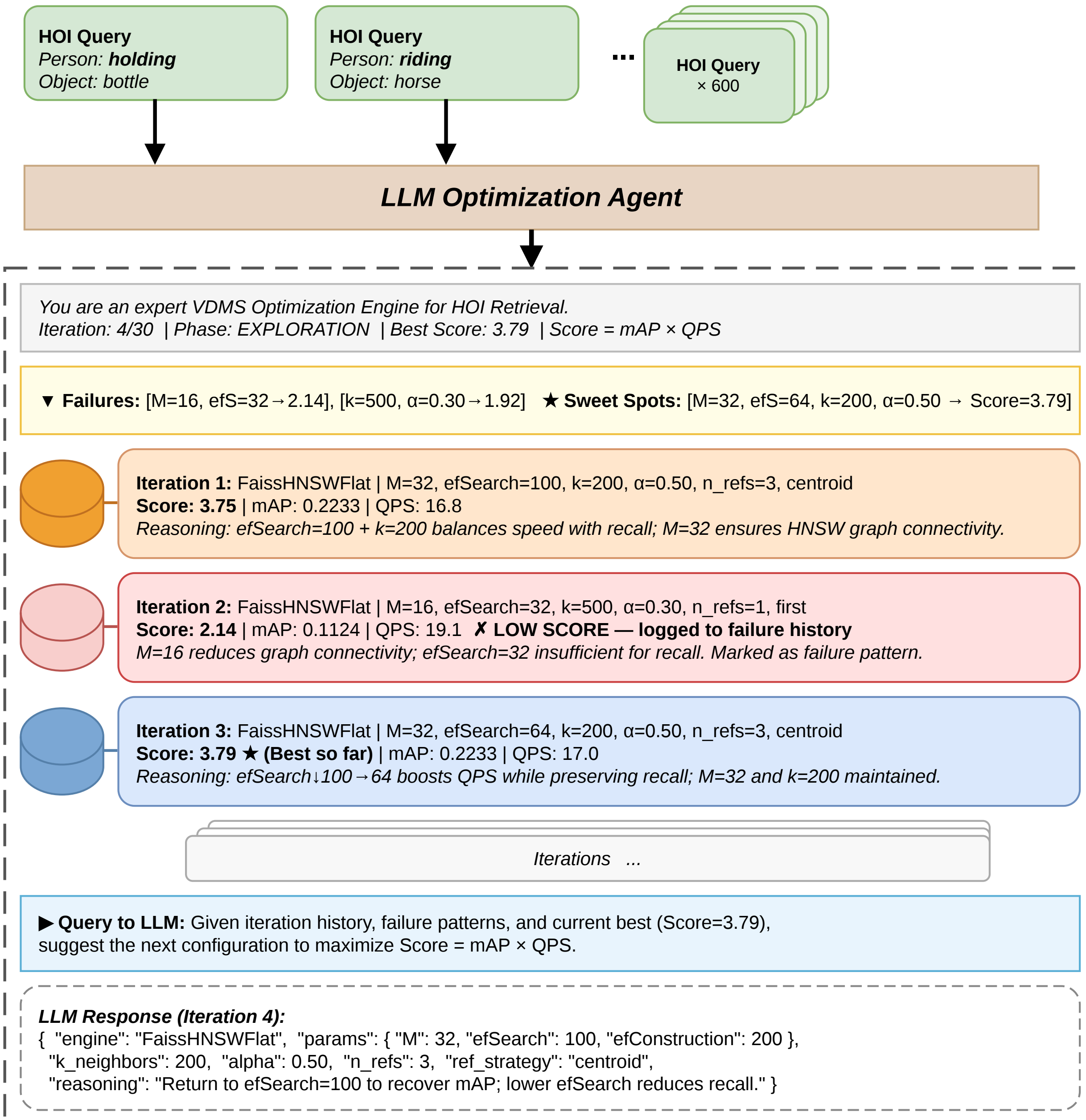

Figure 3: LLM prompt structure. The system prompt encodes parameter physics; the user prompt supplies $g(\mathcal{H}_{t-1})$, history, and untried-value hints, eliciting a structured JSON configuration response.

5.3 Phase-Aware Budget Partition

The N-iteration budget is divided into three phases with boundaries

$$t_{\text{exp}} = \lfloor 0.40\,N \rfloor, \tag{14}$$

$$t_{\text{expl}} = \lfloor 0.75\,N \rfloor, \tag{15}$$

defining **Exploration** ($1 \leq t \leq t_{\text{exp}}$), **Exploitation** ($t_{\text{exp}} < t \leq t_{\text{expl}}$), and **Fine-tuning** ($t_{\text{expl}} < t \leq N$) — for N=30: 12, 10, and 8 iterations respectively. The boundaries encode the structural-before-continuous principle: engine and M determine the achievable QPS ceiling and must be resolved before α and n_r are refined—a conditional dependency that Optuna TPE, which models each parameter independently via univariate kernel density estimation, cannot exploit.

Algorithm 1 LLM-Guided ANN Optimizer

Require: Configuration space Θ; oracle O; budget N
Ensure: Best configuration θ^* and history $\mathcal{H}_N$
$\mathcal{H}_0 \leftarrow \emptyset$; Score$^* \leftarrow 0$; $t_{\exp} \leftarrow \lfloor 0.40N \rfloor$; $t_{\text{expl}} \leftarrow \lfloor 0.75N \rfloor$
for $t = 1, 2, \ldots, N$ **do**
  $\phi_t \leftarrow \text{Phase}(t, t_{\exp}, t_{\text{expl}})$ ▷ EXPLORATION / EXPLOITATION / FINE-TUNING
  $g_t \leftarrow g(\mathcal{H}_{t-1})$ ▷ Eq. (13)
  $\mathcal{U}_t \leftarrow \text{UntriedHints}(\mathcal{H}_{t-1}, \phi_t)$ ▷ $\emptyset$ during EXPLORATION
  $p_t \leftarrow \text{BuildPrompt}(\phi_t, g_t, \mathcal{H}_{t-1}, \mathcal{U}_t)$
  $\theta_t \leftarrow \text{LLM}(p_t)$ ▷ structured JSON response
  if $\theta_t \in \{\theta_s : s < t\}$ **then** ▷ duplicate check
    $\theta_t \leftarrow \text{Fallback}(\Theta \setminus \{\theta_s : s < t\})$
  end if
  $(\text{mAP}_t, \text{QPS}_t) \leftarrow O(\theta_t)$
  $\mathcal{H}_t \leftarrow \mathcal{H}_{t-1} \cup \{(\theta_t, \text{mAP}_t, \text{QPS}_t)\}$
  if $\text{mAP}_t \times \text{QPS}_t > \text{Score}^*$ **then**
    $\text{Score}^* \leftarrow \text{mAP}_t \times \text{QPS}_t$; $\theta^* \leftarrow \theta_t$
  end if
end for
return θ^*, $\mathcal{H}_N$

During Exploration, the *excellent* directive instructs the agent to continue sampling structurally distinct (engine, M, k) combinations rather than converging prematurely; the $\mathcal{U}_t$ hint set is suppressed. During Exploitation, $\mathcal{U}_t$ is activated and *excellent* shifts to reinforcing the best engine region while broadening α–n_r coverage. During Fine-tuning, hints narrow to unexplored reference-strategy variants and proposals are restricted to coordinate-wise perturbations of the current best.

Anti-collapse and stagnation constraints. Let c_t count consecutive proposals in which only α changed relative to the preceding configuration, and let ι_t count iterations elapsed without improvement to Score^*_t. Two hard rules are encoded in the system prompt: (1) if $c_t > 2$, the next proposal must differ structurally—by changing engine, M, k, or n_r—before any further α refinement; (2) if ι_t exceeds a phase-dependent threshold, a stagnation escape directive overrides the current phase instruction and reverts to broad-coverage exploration. Embedding these rules in the prompt rather than as external rejection filters makes structural diversity an explicit part of the agent's reasoning.

Algorithm 1 gives the complete procedure. GP-BO replicates the VDTuner [37] paradigm using optuna.samplers.GPSampler (Gaussian Process surrogate, Expected Improvement acquisition) applied to our single-objective Score. Optuna TPE fits univariate kernel density estimators over each parameter independently. Neither baseline encodes the (k, α) coupling or the engine-conditional structure of Θ; the LLM agent exploits both through its phase-partitioned prompt and guidance function.

6 EXPERIMENTS

6.1 Experimental Setup

Datasets. We evaluate on three benchmarks. **HICO-DET** (Section 3): 47,776 images, 600 compositional text queries, 90,641 ground-truth positive pairs — a stringent testbed combining fine-grained HOI categories, long-tailed relevance distributions, and a cross-modal (text-to-image) retrieval setting. **GLDv2** [35]: Google Landmarks Dataset v2, a large-scale image-to-image retrieval benchmark with 762,000 gallery images and 1,129 public queries spanning global landmarks, providing a 16× larger corpus for cross-domain generalization evaluation. **SIFT1M** [11]: a standard ANN benchmark of one million 128-dimensional SIFT feature vectors with 10,000 queries and exact ground-truth 10-nearest neighbors, providing a complexity-matched control — the index space reduces to a single-stage HNSW problem with no cross-stage coupling — to test whether the LLM's advantage is specific to coupled search spaces. For HICO-DET, image embeddings are 768-dimensional CLIP ViT-L/14 vectors. For GLDv2, each image is represented by a 768-dimensional CLIP ViT-L/14 embedding and a 1024-dimensional DINOv2 ViT-L/14-reg4 embedding, both stored in VDMS. For SIFT1M, raw 128-dimensional SIFT vectors are stored directly in VDMS without any reranking stage; Score = Recall@10 × QPS.

Compared Methods. We compare five optimization strategies, each run for N=30 iterations over Θ:

- **LLM Agent (Ours):** at each iteration, the agent receives the full history $\mathcal{H}_{t-1}$ and a phase-conditioned prompt (Section 5), proposing the next configuration via MiniMax-M2.1 through OpenRouter (3 seeds: 42, 99, 200).
- **GP-BO**: Gaussian Process Bayesian Optimization using optuna.samplers.GPSampler (Expected Improvement acquisition), replicating VDTuner's [37] parameter-independent GP paradigm on our single-objective Score (3 seeds: 42, 99, 200).
- **Optuna TPE** [1]: Tree-structured Parzen Estimator with per-parameter univariate kernel density estimators; adaptive (3 seeds: 42, 99, 200).
- **Random Search** [3]: uniform random sampling over Θ; non-adaptive (3 seeds: 78, 99, 200).
- **Grid Search**: lexicographic enumeration of a fixed 1-D parameter sweep; non-adaptive (2 runs).

As system baselines we include UniIR [34], standalone CLIP ViT-L/14, standalone DINOv2 ViT-L/14-reg4, and CLIP+DINOv2 reranking, all under exact FaissFlat search (Section 6.2).

Metrics. Retrieval quality is measured by mean Average Precision (mAP), computed over all retrieved results per query and averaged across $|Q|$=600 HOI categories, together with Precision@10 and Recall@10 [23]. Throughput is measured in queries per second (QPS), issued as a single batch to the VDMS container. The joint optimization objective $\text{Score} = \text{mAP} \times \text{QPS}$ rewards configurations that simultaneously achieve high retrieval quality and throughput; configurations that sacrifice either axis are penalized multiplicatively. For SIFT1M, where no relevance annotations exist beyond 10-NN ground truth, quality is measured by Recall@10 and the objective is $\text{Score} = \text{Recall@10} \times \text{QPS}$.

Implementation. All experiments run on the NCSA Delta GPU cluster (partition gpuA40x4): one NVIDIA A40 GPU (48 GB VRAM), 16 CPU cores (AMD EPYC 7763), and exclusive node access. VDMS is deployed as an Apptainer container (v2.2.1); all retrieval and optimization code is implemented in Python 3.12. For Optuna, we use the official optuna library [1] with the default TPE sampler. Each optimization iteration builds a complete ANN index from scratch and issues all 600 queries in a single batch, requiring approximately 8–9 minutes of wall-clock time. The full 30-iteration budget therefore demands approximately 3.5–4 hours of compute per method, making sample efficiency — how quickly each method reaches a high-scoring configuration — a practically meaningful criterion. We fix N=30 for all methods and datasets for three reasons. First, 30 iterations represent under 1% coverage of the valid configuration space (e.g., 5,940 HNSW configs on SIFT1M), placing all methods in a regime where optimizer choice is genuinely consequential rather than trivially solvable by exhaustive enumeration. Second, 3.5–4 hours of wall-clock tuning time per method is a realistic practitioner budget; extending to N=100 would require over 12 hours per method without changing the ranking, since our convergence analysis (Figure 4) shows the LLM agent's advantage is established by iteration 10 and the gap only widens thereafter. Third, N=30 is consistent with the evaluation protocol of closely related LLM-guided and Bayesian optimization studies [14, 36].

6.2 System Baselines

Table 3 reports exact-search system baselines that isolate retrieval accuracy from ANN approximation error, establishing upper bounds for each modeling choice.

Compositional difficulty. Even with exhaustive nearest-neighbor search and DINOv2 reranking, the best achievable mAP is only 0.276 (CLIP+DINOv2, FaissFlat, k=2000). This ceiling reflects the structural difficulty of fine-grained HOI categories: classes such as *ride horse*, *sit_on horse*, and *straddle horse* share the same object but differ only in the action verb, and CLIP's joint embedding space partially conflates these distinctions.

UniIR comparison. To provide a fair, apples-to-apples comparison, we evaluate all four UniIR variants (CLIP-SF and BLIP-FF, image-only and multimodal) through the same VDMS FaissFlat index using the identical sequential client-server QPS measurement protocol. UniIR's fine-tuned CLIP-SF achieves mAP = 0.095 in VDMS — a 2.6× drop from the standard CLIP ViT-L/14 zero-shot retrieval (mAP = 0.246, brute-force). This degradation arises because UniIR's multi-task fine-tuning on COCO-style image-text pairs disrupts the precise HOI semantic alignment that the original CLIP text encoder provides for compositional queries such as *"a person ride a motorcycle"*. The BLIP-FF variants perform similarly (mAP ≈ 0.089), confirming that the bottleneck is task-specific semantic alignment rather than model architecture. Despite higher QPS (≈19.7) due to their smaller k=10 search, all UniIR variants achieve Score ≤ 1.871, below our unoptimized CLIP-only baseline (Score = 1.707 at k=500, QPS = 8.21) once mAP differences are accounted for, and far below the LLM-optimized result reported in Section 6.3.

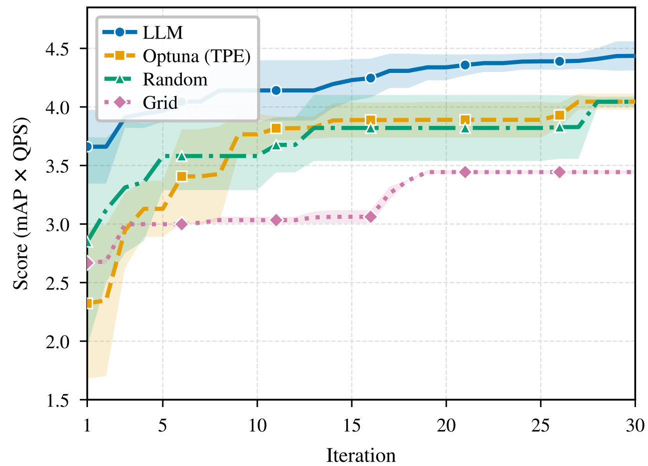

Figure 4: Best Score (mAP×QPS) vs. iteration on HICO-DET (mean ±1σ, 3 seeds; 2 runs for Grid). LLM reaches Score ≈ 3.97 at iteration 6; Optuna TPE at iteration 27, Random Search at iteration 28.

Modality gap. DINOv2 alone (α=1.0, k=2000, mAP = 0.217) underperforms CLIP alone at the same k (mAP = 0.240) by 2.3 percentage points, confirming that language-aligned text embeddings are essential for compositional HOI retrieval. Visual similarity alone fails to discriminate action-level distinctions between HOI categories that share the same object. However, the two modalities are complementary rather than competing: weighted fusion at α=0.5 (k=2000) achieves mAP = 0.276, a gain of +3.65 pp over CLIP alone (+15.2% relative) at negligible latency cost — DINOv2 reranking adds only 4.82 ms over an 855 ms CLIP retrieval pass, a 0.6% overhead. This complementarity motivates including α as a primary optimization target and validates the two-stage pipeline design.

Accuracy–throughput trade-off. Reducing k from 2000 to 500 in CLIP-only retrieval (FaissFlat, α=0) raises QPS by 7.0× (8.21 vs. 1.17) at the cost of only 3.2 percentage points of mAP (0.208 vs. 0.240). The disproportionate Score gain (1.707 vs. 0.280, a 6.1× increase) illustrates a fundamental asymmetry in the Score objective: throughput gains in the high-QPS regime yield multiplicatively larger Score improvements than equivalent mAP gains, because the derivative ∂Score/∂QPS = mAP exceeds ∂Score/∂mAP = QPS whenever QPS ≫ mAP. This asymmetry is precisely the non-linear trade-off that Score = mAP × QPS encodes, and it cannot be navigated effectively without conditioning each new proposal on the full history of observed outcomes.

6.3 End-to-End Optimization Performance

Table 4 and Figure 4 report the complete HICO-DET results across all five optimization methods. The LLM agent achieves the highest mean Score = 4.44, outperforming every baseline: GP-BO (4.29, +3.3%), Optuna TPE (4.05, +9.7%), Random Search (4.04, +9.7%), and Grid Search (3.44, +28.8%) — all relative to the respective baseline mean. This translates to a 2.5× improvement over the unoptimized CLIP FaissFlat baseline (Score = 1.707) and 13.9× over brute-force CLIP+DINOv2 reranking (Score = 0.319). At its

Table 3: System baselines on HICO-DET (FaissFlat exact search). Score = $\mathrm{mAP} \times \mathrm{QPS}$; α: DINOv2 reranking weight (0=CLIP only, 1=DINOv2 only). Latency in Table 5.

Method	α	k	mAP	P@10	R@10	QPS	Score
UniIR [34] baselines							
Zero-shot CLIP ViT-L/14 (brute-force)	—	—	0.2461	0.3370	0.0967	—	—
CLIP-SF Large, image-only (VDMS)	—	10	0.0951	0.4477	0.1327	19.67	1.871
CLIP-SF Large, multimodal (VDMS)	—	10	0.0894	0.4415	0.1206	19.46	1.740
BLIP-FF Large, image-only (VDMS)	—	10	0.0889	0.4420	0.1213	19.69	1.751
BLIP-FF Large, multimodal (VDMS)	—	10	0.0745	0.4097	0.1019	19.72	1.469
Our CLIP + DINOv2 pipeline (FaissFlat exact)							
CLIP ViT-L/14	0.0	500	0.2079	0.3370	0.0967	8.21	1.707
CLIP ViT-L/14	0.0	2000	0.2398	0.3370	0.0967	1.17	0.280
DINOv2 ViT-L/14-reg4	1.0	2000	0.2165	0.3232	0.0800	1.16	0.250
CLIP + DINOv2 rerank	0.5	2000	**0.2763**	**0.3812**	**0.1138**	1.15	0.319

Table 4: ANN optimizer comparison on HICO-DET (30 iterations, means over 3 seeds; Grid: 2 runs). Score = $\mathrm{mAP} \times \mathrm{QPS}$; *Iters→Best*: first iteration at best score. †Grid plateaus at iter. 19. ‡P@10 not recorded for Grid/GP-BO. ¶GP-BO: GPSampler + Expected Improvement [37]. §QDS: single run, difficulty-weighted objective; scores on standard metric.

Method	Score↑	mAP↑	P@10↑	QPS↑	Iters→Best
Grid Search†‡	3.44	0.169	—	20.42	19/30
Random Search	4.04	0.260	0.569	16.13	28/30
Optuna TPE	4.05	0.256	0.529	15.84	27/30
GP-BO¶‡	4.29	0.230	—	18.68	—
LLM Agent (Ours)	**4.44**	0.259	0.487	17.11	30/30
LLM Agent + QDS (Ours)§	**4.76**	0.250	—	**19.07**	19/30

Table 5: System efficiency on HICO-DET (47K images, 600 queries, 5 runs). *Build(s)*: index build; *Lat(ms)*: end-to-end per-query latency.

Configuration	Engine	k	Build(s)↓	CLIP(ms)↓	DINO(ms)↓	Lat(ms)↓	QPS↑
Exact-search baselines							
CLIP only (α=0)	FaissFlat	500	433.3	121.0	0.03	121.1	8.21
CLIP only (α=0)	FaissFlat	2000	431.2	850.0	0.09	850.1	1.17
DINOv2 only (α=1)	FaissFlat	2000	432.9	851.6	5.18	856.8	1.16
CLIP+DINOv2 (α=0.5)	FaissFlat	2000	430.1	855.2	4.82	860.0	1.15
Optimizer best configurations							
LLM Agent (Ours)	HNSW M=48	200	441.8	58.2	0.49	58.9	16.98
Optuna TPE	HNSW M=32	200	423.3	62.0	0.47	62.6	15.97
Random Search	HNSW M=32	200	444.3	48.3	0.32	48.9	20.47
Grid Search	HNSW	—	~445	—	—	—	20.51

best configuration — FaissHNSWFlat, M=48, efSearch=64, k=200, α=0.80, n_r=10, *centroid* reference strategy — the pipeline achieves mAP = 0.255 and QPS = 16.98 (Table 5): a 14.7× throughput gain over brute-force CLIP+DINOv2 (QPS = 1.15) at a mAP cost of only 2.1 percentage points (0.255 vs. the 0.276 exact-search ceiling).

GP-BO comparison. GP-BO — replicating VDTuner's [37] Gaussian Process paradigm — achieves mean Score = 4.29, the second-best result, and trails the LLM agent by 3.3%. Both converge to FaissHNSWFlat as the optimal engine; the gap stems from GP-BO's parameter-independent Expected Improvement acquisition, which cannot represent the conditional dependency between engine type and valid parameter ranges. This structural misspecification manifests as high variance: GP-BO best-scores span 3.93–4.54 across seeds (a 14.4% range), against the LLM agent's 4.34–4.61 (6.2%) — tighter convergence enabled by phase transitions that resolve engine and graph-connectivity choices before entering the continuous (α, n_r) regime.

Convergence and sample efficiency. Figure 4 plots the best-Score-so-far mean $\pm 1\sigma$ curves. The LLM agent's mean curve crosses Score = 3.97 at iteration 6, already exceeding Grid Search's final mean (3.44) and substantially ahead of both Optuna TPE (mean

= 3.41 at iteration 6) and Random Search (3.58 at iteration 6). Optuna TPE's mean curve first matches the LLM's iteration-6 threshold at iteration 27; Random Search at iteration 28. At ≈8–9 minutes per oracle call, this represents concrete practitioner savings: the LLM agent delivers Score ≥ 3.97 approximately 2.7 hours ahead of either adaptive baseline — a gain of 21–22 evaluations within a 3.5–4 hour total budget. This efficiency stems from the phase-aware schedule that resolves structural choices (engine, M) before continuous choices (α, n_r) — a conditional dependency that neither univariate density estimators nor blind sampling can exploit.

Non-obvious operating point. The LLM agent's optimal configuration discloses an operating region inaccessible to parameter-independent methods within 30 iterations. Setting efSearch=64 ≪ k=200 deliberately violates the standard HNSW heuristic of efSearch ≥ k [22], accepting Stage-1 recall loss in exchange for throughput. The loss is recouped at Stage 2 by raising α to 0.80, weighting DINOv2 visual similarity heavily enough to recover relevant neighbors missed by the aggressive ANN approximation. Concurrently, n_r=10 centroid-proximal reference images produce a stable reference vector that minimizes the bias term ε_j^2 in Eq. (10), consistently outperforming n_r=1 across the long-tailed HOI categories where any single image is displaced from the class centroid. The joint (efSearch, k, α) optimum requires modeling the pipeline as a coupled system — a structural dependency that Grid Search's fixed sweep, Optuna TPE's univariate marginals, and Random Search's parameter independence cannot encode.

Failure modes. Grid Search plateaus at iteration 19 with mAP = 0.169 — below the CLIP-only exact-search baseline (mAP = 0.240 at k=2000) — because its lexicographic sweep fixes M=32 and α=0.30, never probing M=48 or α=0.80. Optuna TPE exhibits a structurally distinct failure: at iterations 11–13, all three seeds produce configurations with mAP = 0.169 and QPS = 18.6, a multi-seed plateau caused by TPE's univariate kernel density estimator repeatedly sampling a FaissIVFFlat region it cannot discount without a joint conditional model over engine type. TPE's mean curve first matches the LLM's iteration-6 threshold at iteration 27 — four iterations after the LLM agent has already committed to FaissHNSWFlat and begun refining α and n_r.

6.4 Cross-Domain Generalization: GLDv2 Landmark Retrieval

To test whether the LLM agent's advantage generalizes beyond the HICO-DET benchmark, we evaluate all four optimization methods on the Google Landmarks Dataset v2 (GLDv2) [35], a large-scale image-to-image retrieval benchmark with 762,000 gallery images and 1,129 public queries spanning global landmarks. GLDv2 differs from HICO-DET in two key respects: the corpus is 16× larger (762K vs. 47K images), stressing ANN index construction time and memory, and the retrieval task is image-to-image rather than text-to-image, removing the text-query centroid advantage described in Section 4. Each gallery image is represented by a 768-dimensional CLIP ViT-L/14 embedding and a 1024-dimensional DINOv2 ViT-L/14-reg4 embedding stored in VDMS; the same Score = mAP × QPS objective and 30-iteration budget are used.

Table 6: Optimizer comparison on GLDv2 (762K gallery, 1,129 queries, 30 iterations, seed 42). Score = mAP × QPS. *Iters→Best*: first iteration achieving the reported best score.

Method	Score↑	mAP↑	P@10↑	QPS↑	Iters→Best
Optuna TPE	3.5959	0.1618	0.1968	22.22	23/30
Random Search	3.6491	0.1784	0.2131	20.45	8/30
Grid Search	3.6566	0.1789	0.2125	20.43	21/30
LLM Agent (Ours)	**3.6567**	**0.1790**	**0.2126**	**20.43**	**15/30**

Overall performance. Table 6 reports the GLDv2 results. The LLM agent achieves Score = **3.657**, the best result across all methods, marginally outperforming Grid Search (Score = 3.657), Random Search (Score = 3.649), and Optuna TPE (Score = 3.596). The LLM agent ranks first on both datasets despite the 16× scale increase and the shift from text-to-image to image-to-image retrieval; however, the performance gap between methods is substantially compressed relative to HICO-DET (≲1.7% spread vs. 9.7%), a pattern we return to in Section 6.5.

Scale effects. At 762K images each optimization iteration completes in approximately 5 minutes, with HNSW index construction accounting for ≈272 s of that budget. Because HICO-DET incurs an additional per-query HOI-category overhead absent from image-to-image retrieval, GLDv2 per-iteration wall-clock time is slightly *lower* than HICO-DET (≈5 min vs. 8–9 min), and all 30-iteration budgets complete in ≈2.5 hours per method.

Sample efficiency. The LLM agent reaches its best Score at iteration 15; Grid Search at iteration 21, and Optuna TPE at iteration 23. Random Search locates a competitive configuration at iteration 8, ahead of the LLM agent, reflecting the compressed performance landscape at 762K scale where high-QPS HNSW configurations are easily discovered without conditioning on empirical history. Despite converging later than Random Search, the LLM agent identifies a better final Score (3.657 vs. 3.649), and reaches a Score exceeding all baselines' final performance within 15 iterations.

6.5 Complexity-Matched Validation: SIFT1M

To test whether the LLM agent's advantage is specific to search spaces with cross-stage coupling, we evaluate four methods (LLM Agent, Optuna TPE, Grid Search, Random Search) on SIFT1M [11] under the same 30-iteration budget; GP-BO is omitted from this ablation as it targets the coupled-space comparison on HICO-DET and GLDv2. Unlike HICO-DET, SIFT1M involves no reranking stage and no fusion weight α: the optimizer tunes only HNSW graph connectivity M, construction depth *efConstruction*, search depth *efSearch*, and candidate pool size k — four parameters with no cross-stage coupling. This reduced, structurally simpler space admits no cross-stage dependencies by construction, providing a clean complexity-matched control.

Overall performance. Table 7 reports the results. All four methods converge to FaissHNSWFlat with *efSearch*$\in \{24, 32\}$ as the optimal operating point, yielding final Scores of 1036.0 (LLM), 1035.0 (Grid), 1021.1 (Optuna), and 1014.2 (Random). The spread across all four methods is only 2.1% — a stark contrast to HICO-DET,

Table 7: Optimizer comparison on SIFT1M (1M SIFT vectors, 10,000 queries, 30 iterations, seed 42). Score = Recall@10 × QPS. *Iters→Best*: first iteration achieving the reported best score.

Method	Score↑	Recall@10↑	QPS↑	Iters→Best
Random Search	1014.2	0.912	1111.6	23/30
Optuna TPE	1021.1	0.888	1150.0	14/30
Grid Search	1035.0	0.952	1087.1	25/30
LLM Agent (Ours)	**1036.0**	**0.941**	**1100.8**	**17/30**

where the LLM leads by +9.7% over Optuna TPE, +9.7% over Random Search, and +3.3% over GP-BO (all figures relative to the respective baseline mean Score).

Complexity-proportional advantage. The result confirms the paper's central theoretical prediction: the LLM agent's advantage is complexity-proportional. On HICO-DET, the 7-parameter coupled space (efSearch–k–α interaction across stages) rewards history-conditioned architectural reasoning; removing the cross-stage coupling, as on SIFT1M, eliminates the advantage because the residual 4-parameter ANN search space (engine, M, efSearch, k) is shallow enough for random and grid search to locate the optimum within the same 30-iteration budget. This pattern is consistent with the abstract's theoretical prediction that the agent's benefit scales with the coupling complexity of the configuration landscape.

Sample efficiency. The LLM agent reaches Score $\geq$ 1000 from iteration 1, while Grid Search requires 4 iterations, Optuna TPE requires 11, and Random Search requires 23. Measured by *Iters→Best*, the ranking *inverts* relative to HICO-DET: Optuna TPE reaches its best Score at iteration 14, the LLM agent at iteration 17, Grid Search at iteration 25, and Random Search at iteration 23. This inversion reinforces the complexity-proportional thesis: on the uncoupled 4-parameter space, TPE's univariate acquisition surface is sufficient, and the LLM's architectural history provides no additional navigational advantage. None of the methods exhibits the premature convergence plateau observed in HICO-DET's Optuna and Grid runs, consistent with the simpler landscape having no deep local optima.

6.6 Ablation Study

We isolate the contribution of individual pipeline decisions by comparing performance across systematically varied configurations, using exact-search baselines (Table 3) to control for ANN approximation.

DINOv2 reranking weight (α). Table 3 provides a controlled three-point ablation of α under FaissFlat exact search at k=2000, where approximation error does not confound the comparison. At α=0.0 (CLIP alone), mAP = 0.240. At α=0.5, mAP rises to 0.276 (+3.65 pp, +15.2% relative), with QPS essentially unchanged at 1.15 vs. 1.17 — a near-free accuracy improvement. At α=1.0 (DINOv2 alone), mAP falls to 0.217, confirming that the visual encoder cannot resolve action-level distinctions without language-aligned Stage-1 retrieval. Together, these three points establish that $\alpha \in (0, 1)$ strictly dominates both pure-CLIP and pure-DINOv2 retrieval, and that the optimal α under exact search lies near 0.5. In the ANN setting, the LLM agent discovers α=0.80 — substantially higher than the exact-search optimum — a shift consistent with the reranker serving as an *error corrector* for Stage-1 approximation: at efSearch=64 < k=200, HNSW necessarily misses some relevant neighbors, and increasing α compensates by weighting fine-grained visual reranking more heavily over the returned pool.

Candidate pool size (k). Pool size k controls how many items the DINOv2 reranker sees, trading Stage-1 recall coverage against Stage-2 reranking latency. From Table 3, increasing k from 500 to 2000 at α=0 (FaissFlat) improves mAP from 0.208 to 0.240 (+3.2 pp) but reduces QPS from 8.21 to 1.17 (7.0× throughput loss), yielding a net Score reduction from 1.707 to 0.280. The Score objective therefore strongly favors smaller k: each additional candidate examined adds roughly equal latency whether or not it is relevant, while the mAP gains from a larger pool are subject to diminishing returns as k already covers the majority of relevant items. The LLM agent selects k=200 — smaller than any of the exact-search baseline values — on the reasoning that a larger reference set (n_r=10) and higher reranking weight (α=0.80) compensate for the narrower initial pool, maintaining mAP = 0.255 while preserving the throughput advantage that Score rewards.

Reference selection strategy and count (n_r, ref_strategy). The DINOv2 reranker constructs a class reference vector from n_r corpus images via one of three strategies: *first* (top-1 CLIP-ranked image), *centroid* (mean DINOv2 embedding of the n_r images closest to the class centroid), and *diverse* (greedy farthest-point selection for maximum within-class spread). The LLM agent converges to *centroid* with n_r=10, consistent with the geometric analysis in Section 4: the centroid strategy explicitly minimizes the expected displacement ε_j between the reference vector and the HOI class mean, reducing the reranking bias term ε_j^2 in Eq. (10). The *first* strategy depends on a single CLIP-ranked image that may lie arbitrarily far from the class mean — particularly harmful for the long-tailed categories that dominate HICO-DET. The *diverse* strategy maximizes spread at the cost of mean proximity, yielding a reference vector that spans the class but does not minimize bias. Averaging n_r=10 centroid-proximal images produces a reference that is simultaneously low-bias and low-variance, making it the most reliable proxy for the category distribution.

ANN engine selection. Both the LLM agent and Optuna TPE converge to FaissHNSWFlat as the optimal engine. HNSW's graph-based traversal achieves QPS = 17.11–15.84 for the two best configurations at mAP = 0.259–0.256, versus Grid Search's FaissHNSWFlat/IVFFlat configurations at mean QPS = 20.42 but mAP = 0.169. The Score contrast is stark: 4.44 and 4.05 for HNSW-based methods vs. 3.44 for Grid's sweep — confirming that high-QPS, low-recall configurations are penalized by the Score objective whenever mAP drops below a critical threshold. HNSW's navigable graph structure, with graph connectivity M=48 tuned to maintain sufficient Stage-1 recall, provides the best balance between ANN approximation quality and query throughput for 768-dimensional CLIP embeddings at k=200.

Query Difficulty-Weighted Optimization (QDS). The standard Score = mAP × QPS objective weights all queries equally, yet HOI queries vary substantially in retrieval difficulty: categories with high visual diversity (e.g., *ride horse*) span a wider region of CLIP space than visually compact categories (e.g., *hold cup*), making per-query AP more sensitive to index configuration. We introduce a *Query Difficulty Score* (QDS) that identifies structurally hard queries and upweights their contribution to the optimization objective. For query j, QDS combines two signals. *Gallery entropy* $H_j = -\sum_i p_{ji} \log p_{ji}$ ($p_{ji} = \mathrm{softmax}(\mathrm{f}_{q_j}^{\top}\mathrm{f}_{d_i}/\tau)$) measures how uniformly CLIP mass spreads over the gallery. *Centroid displacement* $\mathrm{CPR}_j = \|\mathrm{f}_{q_j} - \boldsymbol{\mu}_j\|^2/\sigma_j^2$ measures how far the text query lies from the visual centroid of its positives. Normalizing and combining gives $\mathrm{QDS}_j = \tilde{H}_j - \widetilde{\mathrm{CPR}}_j$ with per-query weight $w_j = 1 + \tilde{I}_j \in [1, 2]$ and objective $\mathrm{QDS\text{-}mAP} = \sum_j w_j \cdot \mathrm{AP}_j / \sum_j w_j$.

When the LLM agent optimizes QDS-mAP × QPS, it discovers a configuration (FaissHNSWFlat, M=16, efSearch=500, k=150, α=0.75, n_r=10, *first*) that achieves Score = 4.76 on the *standard* unweighted metric — a +7.3% improvement over the LLM agent with standard objective (Score = 4.44) and the highest single-run Score across all methods on HICO-DET (Table 4). The QDS-guided optimizer reaches its best configuration at iteration 19, suggesting that difficulty-aware guidance accelerates convergence to a better optimum. Notably, the discovered configuration achieves higher QPS (19.07 vs. 17.11) by operating in a regime where the difficulty-weighted signal prefers configurations that maintain precision on hard queries rather than maximizing average recall.

6.7 Discussion

Theory-to-experiment alignment. Section 4 predicts that CLIP text queries should systematically outperform single reference image queries for HOI retrieval, because the text embedding lies near the visual centroid of the HOI class while any individual image is displaced by $\varepsilon_j > 0$. The system baselines in Table 3 provide a controlled empirical test: at identical k=2000 and FaissFlat exact search, CLIP text queries achieve mAP = 0.240 while DINOv2 image-based retrieval achieves mAP = 0.217 (−2.3 pp). This 9.6% relative deficit in favor of text is consistent with the geometric prediction that image queries incur a positive bias $\varepsilon_j^2 > 0$ that text queries avoid. The advantage is expected to be largest for visually diverse HOI categories (large ε_j), where any single reference image is a poor proxy for the class distribution — precisely the setting that dominates HICO-DET's long-tailed structure.

LLM architectural reasoning. Beyond sample efficiency, the LLM agent demonstrates a qualitatively different form of reasoning from Bayesian and non-adaptive baselines: it explicitly reasons about the interaction structure of the pipeline before proposing configurations. During the exploitation phase, the agent selects efSearch=64 < k=200 — a configuration that deliberately accepts ANN recall loss — while simultaneously compensating by raising α to 0.80. This cross-stage compensatory reasoning, in which a relaxed upstream constraint is offset by a tighter downstream parameter, requires modeling the pipeline as an integrated system rather than a product of independent components. Optuna TPE's univariate marginal density model and Random Search's independence assumption are both structurally incapable of encoding this dependency, explaining why both methods identify high-QPS configurations with weaker reranking (QPS = 20.47, mAP = 0.200 for Random's best) rather than the balanced operating point the LLM discovers.

Limitations. The present evaluation spans three datasets (HICO-DET at 47K, GLDv2 at 762K, SIFT1M at 1M vectors) covering both semantic and non-semantic retrieval, but all corpora are considerably smaller than production-grade vector databases at billion scale. Generalization to billion-scale corpora, where memory locality, IVFFlat's coarser quantization effects, and network storage latency introduce coupling dimensions not encountered in the present experiments, remains an open question. Statistical confidence across seeds has been quantified for LLM, Optuna TPE, and Random Search (3 seeds each); extending this to Grid Search and additional LLM families remains future work. Finally, the LLM agent's behavior is specific to MiniMax-M2.1 via OpenRouter; different model families may differ in convergence speed, architectural priors, and sensitivity to the five feedback class labels used in the per-iteration diagnostic.

7 CONCLUSION

Standard parameter-independent HPO methods fail structurally on two-stage ANN retrieval pipelines because engine-type gating, search-depth–fusion-weight cross-stage coupling, and the latency–recall trade-off form conditional dependencies that univariate acquisition surfaces cannot model. We presented a phase-aware LLM optimization agent that conditions each configuration proposal on the full trajectory $\mathcal{H}_t$ of evaluated (configuration, Score) pairs, partitions its 30-iteration budget into exploration, exploitation, and refinement phases, and enforces an anti-collapse constraint to prevent stagnation in high-scoring but structurally similar regions. On HICO-DET — the most compositionally coupled of our three benchmarks — the agent achieves mean Score = 4.44, reaching Score ≥ 3.97 at iteration 6, a threshold that requires 27 iterations for Optuna TPE and 28 for Random Search (≈2.7 hours of compute saved at 8–9 minutes per oracle call). The +9.7% advantage over both Optuna TPE and Random Search, and +3.3% over GP-BO, compresses to ≲1.7% on GLDv2 and 2.1% on SIFT1M as coupling complexity decreases — with the SIFT1M convergence ordering inverting to favor Optuna TPE (iteration 14) over the LLM agent (iteration 17) — confirming that the agent's benefit scales proportionally with configuration-landscape coupling complexity.

Three systems-level extensions follow directly from this framing. First, billion-scale corpora introduce memory-topology coupling between HNSW graph partitioning and NUMA-node locality; extending the agent's context with hardware-profile telemetry is necessary to navigate these additional conditional dimensions without exhausting the per-iteration budget in construction overhead alone. Second, multi-tenant workloads introduce query-mix heterogeneity across the index lifetime; the agent must condition proposals on observed workload-distribution shifts rather than a static performance trajectory $\mathcal{H}_t$, requiring an online history-weighting scheme that discounts stale evaluations. Third, replacing the offline oracle $O(\theta)$ with closed-loop feedback from a live VDMS execution engine — where streaming QPS measurements update $\mathcal{H}_t$ continuously — would enable re-optimization without full index rebuilds, eliminating the dominant 8–9-minute per-call wall-clock cost that currently bounds the practical iteration budget.

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